

Benchmarking Factual Accuracy of Large Language Models Across Scientific Domains

A Review and Synthesis of Evaluation Methodologies, Benchmarks, and Open Problems

Abstract

Large language models (LLMs) are increasingly consulted for scientific information spanning medicine, chemistry, physics, and biology, yet their propensity to generate fluent but factually incorrect content — commonly termed hallucination — poses acute risks in these high-stakes domains (Huang et al., 2025; Ji et al., 2023). This paper surveys the landscape of benchmarks and methodologies developed to measure the factual accuracy of LLMs across scientific fields. We trace the evolution of evaluation paradigms from general-purpose truthfulness probes such as TruthfulQA (Lin, Hilton, & Evans, 2022) and multitask knowledge tests such as MMLU (Hendrycks et al., 2021) to domain-specialized instruments including MultiMedQA and Med-PaLM for clinical knowledge (Singhal et al., 2023), ChemBench for chemistry (Mirza et al., 2025), GPQA for graduate-level physics, biology, and chemistry (Rein et al., 2023), and SciFact for biomedical claim verification (Wadden et al., 2020). We examine current approaches to factuality measurement, including atomic fact decomposition (Min et al., 2023), short-form adversarial question answering (Wei et al., 2024), and retrieval-augmented verification. We then discuss persistent challenges — benchmark contamination, saturation, single-answer assumptions, and the high cost of expert annotation — before outlining research gaps in dynamic, multilingual, and multimodal scientific factuality evaluation. The review concludes that while domain-specific benchmarks have substantially improved the granularity of factuality assessment, no existing framework yet provides continuous, contamination-resistant, and cross-disciplinary coverage of scientific knowledge, underscoring the need for coordinated, living benchmark infrastructures.

Keywords: *large language models; factual accuracy; hallucination; benchmarking; scientific domains; question answering; knowledge evaluation; retrieval-augmented generation*

1. Introduction

The rapid integration of large language models into scientific workflows — literature summarization, clinical decision support, chemical property prediction, and general science communication — has made the factual reliability of these systems a central concern for researchers, clinicians, educators, and policymakers. Unlike stylistic or grammatical errors, factual inaccuracies in scientific contexts carry outsized consequences: a hallucinated drug interaction, an incorrect chemical reactivity claim, or a fabricated citation can mislead downstream decisions in medicine, materials design, or public communication (Huang et al., 2025). Because LLMs generate text through next-token prediction over patterns learned from large, heterogeneous corpora rather than through grounded symbolic reasoning over verified facts, they are structurally prone to producing statements that are fluent, plausible, and wrong (Ji et al., 2023).

Benchmarking factual accuracy is therefore not a peripheral evaluation exercise but a prerequisite for responsible deployment. Early efforts to quantify LLM truthfulness, such as TruthfulQA (Lin, Hilton, & Evans, 2022), demonstrated that scaling model size alone does not guarantee truthfulness and can even amplify imitative falsehoods absorbed from training data. Subsequent multitask benchmarks such as MMLU (Hendrycks et al., 2021) broadened evaluation to dozens of academic subjects, including substantial STEM content, revealing that models could achieve credible aggregate scores while still failing on narrow, expert-level questions. These general-purpose instruments, however, were not designed to isolate factuality specifically within scientific reasoning, nor to resist the possibility that evaluation items were memorized during pretraining.

This has motivated a wave of domain-specific benchmarks purpose-built for scientific factuality: biomedical question answering suites such as MultiMedQA (Singhal et al., 2023), chemistry-specific evaluation frameworks such as ChemBench (Mirza et al., 2025), Google-proof graduate-level question sets in physics, biology, and chemistry such as GPQA (Rein et al., 2023), and claim-verification datasets grounded directly in the scientific literature such as SciFact (Wadden et al., 2020). In parallel, methodological innovations — atomic fact decomposition (Min et al., 2023), adversarially collected short-answer factuality probes (Wei et al., 2024), and retrieval-augmented verification pipelines — have refined how factuality itself is measured, moving beyond binary correctness toward fine-grained precision and calibration.

This paper synthesizes this fast-moving literature into a structured overview. Section 2 reviews the background and prior work on hallucination and factuality evaluation. Section 3 develops a technical discussion of how scientific-domain benchmarks are constructed and scored. Section 4 surveys current benchmarks and mitigation approaches organized by scientific domain. Section 5 critically examines the limitations of the present evaluation ecosystem. Section 6 identifies open research gaps and promising directions, and Section 7 concludes.

2. Background and Related Work

2.1 Hallucination and Factuality as Distinct but Related Constructs

The terms "hallucination" and "factuality" are closely related but not identical. Hallucination generally refers to generated content that is unfaithful to a source (intrinsic hallucination) or unsupported by any verifiable ground truth (extrinsic hallucination), whereas factuality is a broader property describing the correspondence between a model's output and established facts about the world, irrespective of a specific source document (Ji et al., 2023; Huang et al., 2025). Huang et al. (2025) provide a comprehensive taxonomy distinguishing factuality hallucination — inconsistency with verifiable real-world facts — from faithfulness hallucination — divergence from user instructions or provided context — and catalogue detection and mitigation techniques spanning retrieval augmentation, decoding-time intervention, and post-hoc correction. Yang et al. (2025) similarly survey factuality along the axes of knowledge storage, retrieval mechanisms, and domain specificity, noting that factual reliability degrades unevenly across knowledge domains

and time horizons. Wang et al. (2024) extend this analysis specifically to the post-2023 generation of instruction-tuned and retrieval-augmented LLMs, cataloguing benchmarks by whether they test discrimination (identifying which of several statements is true) or generation (producing factually sound content unprompted).

2.2 General-Purpose Factuality and Knowledge Benchmarks

Several foundational benchmarks, though not scientific-domain-specific, established the methodological groundwork for later specialized instruments. TruthfulQA (Lin, Hilton, & Evans, 2022) comprises 817 questions across 38 categories, including health and science-adjacent topics, deliberately crafted to elicit imitative falsehoods that mirror common human misconceptions; the original study found that the largest GPT-3 variants were paradoxically less truthful than smaller ones, since larger models more faithfully reproduce popular but false beliefs present in training text. MMLU (Hendrycks et al., 2021) evaluates zero- and few-shot multitask accuracy across 57 subjects spanning STEM, humanities, and professional domains, and has become a near-universal reporting benchmark for frontier models; its successor, MMLU-Pro (Wang et al., 2024, arXiv:2406.01574), increases the answer-option count and question difficulty to reduce the ceiling effects that had begun to saturate the original test. SimpleQA (Wei et al., 2024) narrows the focus to short, single-answer, fact-seeking questions adversarially collected against GPT-4 responses, explicitly rewarding models that abstain when uncertain rather than guessing — a calibration-sensitive design later extended by SimpleQA Verified, which corrects distributional and label-quality issues in the original set.

2.3 From General Knowledge to Scientific Domains

The transition from general knowledge testing to scientific-domain factuality evaluation reflects a recognition that aggregate accuracy on broad benchmarks masks domain-specific failure modes. Medicine was among the first domains to receive dedicated treatment: MultiMedQA (Singhal et al., 2023) aggregates six existing medical question-answering datasets — including MedQA, MedMCQA, and PubMedQA — with a new consumer health query set, HealthSearchQA, and proposes a human evaluation rubric spanning factuality, comprehension, reasoning, possible harm, and bias, moving beyond simple accuracy toward clinically meaningful assessment. Chemistry received comparable treatment more recently through ChemBench (Mirza et al., 2025), which benchmarks LLMs against practicing chemists across more than 2,700 curated question-answer pairs. Cross-disciplinary efforts such as GPQA (Rein et al., 2023) instead target expert-level difficulty directly, constructing questions in biology, physics, and chemistry that remain difficult even for skilled non-experts with unrestricted web access. Scientific claim verification, exemplified by SciFact (Wadden et al., 2020), reframes factuality evaluation as a retrieval-and-entailment task grounded in the primary biomedical literature rather than as multiple-choice recall.

3. Main Thematic and Technical Discussion

3.1 Taxonomy of Evaluation Formats

Scientific factuality benchmarks can be organized along several orthogonal design axes. The first is response format: multiple-choice question answering (MMLU, GPQA, ChemBench) permits low-cost automated scoring but risks inflating apparent competence through elimination heuristics rather than genuine domain reasoning, a concern explicitly raised in chemistry-focused evaluations (Mirza et al., 2025). Short-form open generation (SimpleQA) requires an unambiguous, singular correct answer and is graded by an automated or human rater comparing the generated string against a gold answer. Long-form open generation — for instance, a model asked to summarize a mechanism of drug action or describe a chemical reaction pathway — cannot be scored by simple string matching and instead requires decomposition-based or claim-verification-based scoring, discussed below.

A second axis is the source and difficulty calibration of questions. Some benchmarks, notably GPQA (Rein et al., 2023), are explicitly constructed to be "Google-proof": questions are validated to resist solution via unrestricted web search by highly skilled non-experts, who in the original study achieved only 34% accuracy despite more than 30 minutes of search time per question, compared to 65% for relevant PhD-level experts. This design targets genuine reasoning and deep domain knowledge rather than shallow retrieval, and it directly addresses a core threat to benchmark validity: the possibility that a model has simply memorized the answer to a search-indexed question during pretraining.

3.2 Atomic Fact Decomposition and Fine-Grained Scoring

For long-form scientific text, binary correct/incorrect judgments are inadequate because a single response typically mixes accurate and inaccurate content. FActScore (Min et al., 2023) addresses this by decomposing generated text into "atomic facts" — minimal, independently verifiable propositions — and computing the percentage supported by a trusted knowledge source, yielding a continuous factual-precision metric rather than a pass/fail label. Applied to biography generation, this method revealed that even strong commercial systems such as ChatGPT achieved only 58% factual precision, a gap invisible to coarser evaluation. This atomic-decomposition philosophy has since propagated into scientific claim verification pipelines, including SciFact-style systems that identify SUPPORTS, REFUTES, or NOT-ENOUGH-INFO labels against evidentiary abstracts drawn from peer-reviewed biomedical literature (Wadden et al., 2020).

3.3 Calibration, Abstention, and the Cost of Overconfidence

A recurring technical theme across current benchmarks is the treatment of model confidence and abstention. SimpleQA's grading scheme explicitly separates "correct," "incorrect," and "not attempted" responses and computes an F-score balancing correctness against willingness to answer, since a system that always guesses can inflate raw accuracy while providing unreliable information in practice (Wei et al., 2024). ChemBench similarly probes whether models can estimate the difficulty or reliability of their own answers, finding that confidence estimates

frequently show no meaningful correlation with actual correctness (Mirza et al., 2025) — a finding consistent with broader observations that LLMs tend to be poorly calibrated and often overstate confidence relative to true accuracy, a pattern also identified in the original TruthfulQA study when comparing stated confidence to empirical correctness across model families. This overconfidence is particularly hazardous in scientific and clinical contexts, where a user may reasonably infer that fluent, assertively phrased text reflects verified knowledge.

3.4 Retrieval Augmentation as Both Mitigation and Confound

Retrieval-augmented generation, in which a model consults external documents before answering, is widely used both to mitigate factual errors and, paradoxically, to complicate benchmark design. When tool use or search access is permitted, short-form factuality benchmarks such as SimpleQA Verified report performance approaching ceiling, which is informative about retrieval efficacy but uninformative about a model's parametric — internally stored — knowledge; consequently, benchmark designers must explicitly control whether tools are enabled and report results separately for parametric and retrieval-augmented settings. In biomedical contexts, retrieval-augmented frameworks that draw on curated corpora such as PubMedQA, MedMCQA, and BioASQ are used both as training signal and as evaluation infrastructure, allowing model outputs to be cross-validated against primary literature rather than against the model's own generative fluency.

4. Current Approaches and Developments

4.1 Medicine and Clinical Knowledge

The medical domain has the most mature benchmarking ecosystem. MultiMedQA (Singhal et al., 2023) combines MedQA (US Medical Licensing Examination-style questions), MedMCQA, PubMedQA, LiveQA, MedicationQA, and MMLU's clinical subsets with the newly introduced HealthSearchQA. Evaluating PaLM and its instruction-tuned variant Flan-PaLM on this suite, the authors report that Flan-PaLM achieved 67.6% accuracy on MedQA, exceeding prior state of the art by more than seventeen percentage points, yet human evaluation using a rubric spanning factuality, comprehension, reasoning, possible harm, and bias revealed persistent gaps relative to practicing clinicians. This gap motivated instruction prompt tuning and the resulting Med-PaLM model, illustrating a broader pattern in the field: aggregate multiple-choice accuracy can substantially overstate real-world clinical readiness, and multi-axis human evaluation frameworks remain necessary to surface safety-relevant shortcomings that automated scoring cannot detect.

4.2 Chemistry

ChemBench (Mirza et al., 2025) provides an automated framework spanning more than 2,700 curated question–answer pairs across general, inorganic, organic, and physical chemistry, materials science, technical chemistry, and toxicity and safety. The study benchmarked leading open- and closed-source LLMs directly against practicing chemists and found that the strongest

models outperformed the best human participants on average, yet still failed basic tasks, exhibited overconfident predictions, and showed no reliable correlation between molecular structural complexity and answer accuracy — evidence, the authors argue, that strong aggregate performance can coexist with reliance on memorization rather than structural chemical reasoning. Related efforts such as ChemLLM’s ChemBench task suite adopt a parallel multiple-choice methodology explicitly chosen over BLEU/ROUGE-style text-similarity metrics, because stylistic similarity metrics can reward fluent but factually incorrect chemical statements.

4.3 Physics, Biology, and Cross-Disciplinary Graduate-Level Reasoning

GPQA (Rein et al., 2023) targets the intersection of physics, biology, and chemistry with 448 multiple-choice questions written by domain experts and validated for resistance to web search. The strongest GPT-4-based baseline in the original study achieved only 39% accuracy, well below the roughly 65–74% range achieved by relevant PhD-level experts, establishing GPQA and its harder curated subset, GPQA Diamond, as a stress test for models that already saturate easier benchmarks such as MMLU. The explicit goal articulated by the benchmark’s authors is to support scalable oversight: as AI systems approach or exceed human capability on specific scientific questions, benchmarks must be constructed so that human evaluators can still meaningfully judge correctness, or alternative supervision strategies must be developed.

4.4 Scientific Claim Verification and Literature Grounding

A complementary approach evaluates not whether a model can answer a question, but whether it can correctly judge the veracity of a scientific claim against literature evidence. SciFact (Wadden et al., 2020) introduced this task with 1,409 expert-written biomedical claims paired with a corpus of 5,183 research abstracts, each claim labeled as supported, refuted, or not addressed by identified evidentiary sentences. This claim-verification paradigm has since been extended along multiple dimensions: SciFact-Open enlarges the evidence corpus to roughly 500,000 abstracts to better reflect open-domain retrieval difficulty, HealthVer and PubHealth extend coverage to public-health claims from search queries and journalistic fact-checking, and domain-specific successors such as SciTab (computer science, reasoning over tables) and materials-science claim-verification benchmarks demonstrate the generality of the paradigm beyond biomedicine. Because these tasks require retrieving and reasoning over primary literature rather than recalling memorized answers, they are comparatively resistant to the benchmark contamination concerns that affect static multiple-choice tests.

4.5 Automated Long-Form Factuality Metrics

Beyond dataset construction, methodological developments in scoring have materially advanced the field. FActScore (Min et al., 2023) automated its atomic-fact evaluation using a retrieval-plus-LLM pipeline achieving less than a two percent error rate relative to costly human annotation, enabling evaluation of thousands of long-form generations at a fraction of the cost of manual review. Such automated factuality scorers are increasingly used as a bridge between narrow

multiple-choice benchmarks and the free-form, long-form scientific writing that constitutes much real-world LLM usage, including literature summarization, explanation generation, and drafting of research communications.

5. Research Challenges and Limitations

5.1 *Benchmark Contamination and Saturation*

A pervasive threat to the validity of static benchmarks is training-data contamination: because frontier models are pretrained on vast web-scraped corpora, benchmark questions and their answers — particularly for long-standing datasets such as MMLU — may already appear in training data, inflating measured accuracy without reflecting genuine reasoning ability. This concern directly motivated the "Google-proof" design of GPQA (Rein et al., 2023), yet even carefully constructed benchmarks face gradual saturation as newer models are trained on larger and more recent corpora, a dynamic that produced MMLU-Pro as a harder successor to MMLU once ceiling effects emerged on the original test.

5.2 *The Single-Answer Assumption and Ambiguity*

Short-form factuality benchmarks such as SimpleQA deliberately restrict questions to those with a single, indisputable answer to enable reliable automated grading (Wei et al., 2024); the original study estimated an irreducible error rate near 3% even after this careful curation, primarily from residual question ambiguity. This design choice, while methodologically necessary, excludes much of the nuance characteristic of genuine scientific discourse, where correct answers are frequently probabilistic, context-dependent, or subject to legitimate expert disagreement — a mismatch that limits the ecological validity of short-form benchmarks as proxies for real-world scientific communication quality.

5.3 *Cost and Scarcity of Expert Annotation*

High-quality scientific benchmarks require subject-matter experts to write, validate, and periodically refresh questions — a resource-intensive process. GPQA's construction relied on PhD-level domain experts across three disciplines, and MultiMedQA's human evaluation framework required clinician raters assessing responses along five distinct axes (Singhal et al., 2023). ChemBench's comparison against practicing chemists likewise depended on a bespoke expert survey infrastructure (Mirza et al., 2025). This annotation burden constrains both the scale and the update cadence of scientific benchmarks relative to the pace of model development, creating persistent latency between the emergence of new model capabilities and the availability of benchmarks capable of rigorously assessing them.

5.4 *Discrimination-Generation Gap and Multiple-Choice Artifacts*

Multiple-choice formats, while cheap to score, can be solved through elimination or superficial pattern matching rather than genuine recall or reasoning, a limitation explicitly noted in chemistry

benchmarking work (Mirza et al., 2025) and echoed across the broader factuality survey literature (Wang et al., 2024). A model's ability to discriminate between provided answer choices does not necessarily transfer to its ability to generate correct scientific content unprompted, meaning that strong multiple-choice performance can overstate real-world reliability in open-ended scientific writing or clinical documentation tasks.

5.5 Domain and Temporal Coverage Gaps

Existing scientific benchmarks remain unevenly distributed across disciplines: medicine, chemistry, and general physics/biology are comparatively well covered, whereas fields such as earth science, astronomy, and engineering subdisciplines are underrepresented. Nearly all current benchmarks are static snapshots that cannot reflect scientific knowledge that changes after their construction date — a structural mismatch for rapidly evolving fields such as biomedicine, where guidance and consensus can shift within months of new clinical trial results.

6. Research Gaps and Future Directions

Several directions appear promising for advancing the rigor and coverage of scientific factuality evaluation. First, dynamic and continuously refreshed benchmarks — analogous in spirit to time-stamped question-answering platforms — could reduce contamination risk and track model performance against evolving scientific consensus rather than a fixed historical snapshot, an approach only nascently explored outside general-knowledge question answering. Second, deeper integration of claim-verification methodology, as pioneered by SciFact (Wadden et al., 2020) and its successors, into mainstream scientific benchmarking would shift evaluation emphasis from answer recall toward literature-grounded reasoning, better reflecting how scientists themselves verify claims.

Third, the field would benefit from standardized, cross-domain protocols for calibration and abstention reporting. Current practice varies widely: some benchmarks reward confident guessing, others explicitly penalize overconfidence (Wei et al., 2024), and few report calibration curves alongside raw accuracy. A shared reporting standard — for instance, mandatory disclosure of accuracy conditioned on stated confidence, following the spirit of ChemBench's confidence-estimation analysis (Mirza et al., 2025) — would allow more meaningful cross-benchmark and cross-model comparison of trustworthiness rather than raw capability alone.

Fourth, underexplored scientific subfields require dedicated benchmark development. While medicine, chemistry, and core physical sciences now have relatively mature evaluation ecosystems, environmental science, materials engineering beyond chemistry, and interdisciplinary translational research remain comparatively underserved, despite growing LLM use in these areas. Fifth, multilingual and multimodal scientific factuality remains largely open: nearly all benchmarks reviewed here are English-centric and text-only, whereas scientific communication increasingly involves figures, tables, chemical structures, and non-English literature — gaps

partially addressed by emerging multimodal chemistry benchmarks but not yet systematically extended across scientific disciplines.

Finally, closer coupling between factuality benchmarking and mitigation research — retrieval augmentation, fine-tuning on verified corpora, and post-hoc correction methods — would help clarify which interventions most reliably improve factual precision in scientific domains without degrading fluency or helpfulness, and under what conditions retrieval access substitutes for versus complements parametric scientific knowledge.

7. Conclusion

The factual accuracy of large language models in scientific domains has moved from an informally observed weakness to a rigorously benchmarked property, supported by an increasingly diverse ecosystem of evaluation instruments. General-purpose factuality and knowledge tests such as TruthfulQA and MMLU established foundational methodology, while domain-specialized benchmarks — MultiMedQA and Med-PaLM in medicine, ChemBench in chemistry, GPQA across physics, biology, and chemistry, and SciFact for literature-grounded claim verification — have provided the granularity necessary to expose failure modes invisible to aggregate, general-domain scoring. Methodological advances such as atomic fact decomposition and adversarially collected short-form factuality probes have further sharpened what "measuring factuality" actually means in practice, moving the field beyond binary correctness toward calibrated, fine-grained assessment.

Nonetheless, the review identifies persistent structural limitations: benchmark contamination and saturation, the artificial simplicity imposed by single-answer and multiple-choice formats, the high cost and limited scale of expert annotation, and uneven coverage across scientific subfields and languages. Addressing these limitations will require dynamic, continuously updated evaluation infrastructures, standardized calibration reporting, deeper integration of literature-grounded claim verification, and expanded coverage of underserved scientific domains and modalities. As LLMs become further embedded in scientific research, education, and clinical practice, the maturation of factuality benchmarking is not merely an academic exercise but a necessary precondition for the safe and trustworthy use of these systems in consequential, evidence-based domains.

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